

FIN ST447–ST461

Global Gain Transition, Exact Degree-Six Invariant Relations, and Adaptive Infrared Continuation

Release 10.87

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Abstract

Programs ST447–ST461 continue the local analytical and computational FIN research program after Release 10.86. The principal result is the first computer-assisted global lower certificate close to the previously isolated reflection-even coexistence branch. For the supplied dimensionless family

$$V_g(p) = D(p||u) - \frac{g}{2}(p-u)^T A(p-u), \quad p \in \Delta_{11},$$

the uniform state is now proved to be the unique global minimizer for every $0 \leq g \leq 2.8934$. A previously certified nonuniform competitor has negative value at $g = 2.9024964917477667$. Consequently a first global transition exists in the interval

$$[2.8934, 2.9024964917477667],$$

of width 0.0090964917477665. The identity of the first competing orbit and transition uniqueness remain open; in particular, the local branch crossing is not silently promoted to the global transition.

The invariant-algebra result is exact. A two-prime calculation reconstructs sixteen primitive integer relations among the 326 declared decomposable degree-six invariants. Direct characteristic-zero polynomial division by $x_0 + \cdots + x_{11}$ verifies every relation. Together with a nonzero modular rank-310 minor, this proves characteristic-zero decomposable rank 310 and primitive quotient dimension $365 - 310 = 55$. At degree eight, a projective incidence census checks every triple made collinear by two independent finite field projections against the complete evaluation matrix. No rational syzygy of support at most three exists; the forced relation basis must begin at support four or larger.

The compactified infrared branch is continued by 28 overlapping parametric Krawczyk cells from $b = 0.017$ to $b = 0.1$, with independently certified common root boxes at every interface. Combined with ST437, this proves one connected locally unique component from $b = 0$ through 0.1, retaining ordering, orientation, and local Brouwer degree -1 . It does not count roots outside the chain. Secondary programs expose two useful negative results: five synthetic design exchanges improve in-design E/A conditioning but worsen frozen holdout leverage, and larger Ulam grids invalidate the earlier visual plateau interpretation. No program derives the gain, supplies a nonpremise selector, fixes dimensional units, transfers legacy physical roles, or adds independent empirical evidence.

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1 Scope, source packet, and nonclaims

The audited tracked baseline is Git commit `bb540a767e210a5a3bf816e4868e7bcefbac3872`, supplemented by the local ST417–ST446 executable packets listed in Section 9. The strict finite operator is

$$A = sI - W, \quad W_{xx} = 0, \quad W_{xy} = \frac{\cos(0.18575d(x,y) + 0.16250)}{1 + d(x,y)^{9/5}} \quad (x \neq y), \quad (1)$$

on $\mathbb{Z}_{12}$, with off-diagonal row sum $s = 1.660307278766099\dots$. Every quantity in this report is dimensionless.

The legacy ontological kernel remains an intermediate bridge kernel; it is not silently identified with (1). No legacy claim concerning electroweak parameters, electromagnetic coupling, a gravity hierarchy, Planck scales, or matter generation is transferred. The project ontology is retained only as context: the nadsoliton itself is primordial fractal information in a solitonic state, with no informational substrate below it, and with preferred internal order nadsoliton $\rightarrow$ light $\rightarrow$ matter $\rightarrow$ emergent observer.

A global orbit is not a selector. QW-2191 remains open. A theorem conditional on g does not source g . The infrared variables are names in a dimensionless optimization fixture and are not physical wavelengths or energies. Local computation is not external validation and does not establish a Theory of Everything.

2 Executive result ledger

Program	Status	Result
ST447	[PROVEN]	No rational degree-eight relation supported on one, two, or three of the 2028 declared candidates. The forced relation basis remains unknown.
ST448	[PROVEN]	The uniform state is the unique global minimizer for every supplied $0 \leq g \leq 2.8934$.
ST449	[PROVEN]	A first global change exists in $[2.8934, 2.9024964917477667]$; branch identity and uniqueness are open.
ST450	[PROVEN] limited	The partial Morse polynomial factors as $1 + (1+t)(12+30t)$ and satisfies the strong Morse condition. This does not prove completeness.
ST451	[PROVEN] local	Each certified index-one saddle has a stabilizer-invariant one-dimensional unstable eigenspace. Heteroclinic endpoints remain numerical.
ST452	[PROVEN] local	Twenty-eight Krawczyk cells and 28 certified interface boxes continue one component through $b = 0.1$.
ST453	[PROVEN] local	The component has $y_1 < y_2 < y_3$, $0 < a < 1$, $T > 1$, negative Jacobian orientation, and local degree -1 .
ST454	[PROVEN]	Sixteen exact integer relations close the characteristic-zero degree-six rank at 310 and quotient dimension at 55.
ST455	[STRONG NUMERICAL EVIDENCE] / falsified robustness	Five exchanges improve E by 17.20% and A by 15.87%, but worsen frozen holdout leverage; robustness is rejected.
ST456	[STRONG NUMERICAL EVIDENCE] / caution	Grids 71 and 81 show continued Ulam-ratio drift. No infinite-dimensional two-sided transfer-gap bracket exists.
ST457	[PROVEN] conditional	The conservative scalar ISS architecture has a unique optimal radius 1.9360740×10^{-6} .
ST458	[BLOCKED]	Current sign-aware and absolute-value relaxations do not close the rank-seven global problem.
ST459	[BLOCKED]	No internal strict source of g is exported.
ST460	[BLOCKED]	No nonpremise selector is exported; QW-2191 remains open.
ST461	[BLOCKED]	No independent referee, laboratory record, custody chain, or held-out empirical datum is added.

3 ST448–ST449: the first narrow global-transition theorem

Let $u_i = 1/12$, $q = p - u$, and

$$D(p\|u) = \sum_{i=0}^{11} p_i \log(12p_i), \quad V_g(p) = D(p\|u) - \frac{g}{2} q^T A q. \quad (2)$$

Because A is positive semidefinite and has constants as its kernel, $V_g(p)$ is nonincreasing in g for each fixed nonuniform p .

3.1 A separable entropy-curvature lemma

Put $r = 1/12$ and

$$h(m) = \frac{m \log(12m) - (m - r)}{(m - r)^2}. \quad (3)$$

Elementary differentiation shows that

$$\frac{x \log(12x) - (x - r)}{(x - r)^2} \geq h(m) \quad \text{for } 0 \leq x \leq m, \quad m > r.$$

The removable value at $x = r$ is 6. Summing the scalar inequalities and using $\sum_i (p_i - r) = 0$ yields

$$D(p\|u) \geq h(m) \|p - u\|_2^2 \quad \text{whenever } \max_i p_i \leq m. \quad (4)$$

Since $q^T A q \leq \lambda_{\max}(A) \|q\|^2$, the sector is strictly positive if $h(m) > g \lambda_{\max}/2$.

At $g = 2.8934$ and

$$\lambda_{\max}(A) = 2.342182041146300\dots, \quad m_0 = 0.33426904850345385,$$

the paid scalar margin is

$$h(m_0) - \frac{g \lambda_{\max}}{2} = 4.949286539379649 \times 10^{-4} > 0.$$

Thus the complete low-maximum sector is closed analytically.

3.2 The complementary two-variable cover

Select a largest coordinate and write

$$p_0 = 1 - t, \quad p_j = t y_j, \quad \sum_{j=1}^{11} y_j = 1, \quad \rho = \sum_{j=1}^{11} y_j^2.$$

The high sector is the rectangle

$$0 \leq t \leq 1 - m_0, \quad 1/11 \leq \rho \leq 1.$$

At fixed ρ , the entropy minimum is bounded by the one-large, ten-equal extremal distribution. For the quadratic part write $W = sI - A$ and let $R = W_{\{1, \dots, 11\}}$. With $\bar{y} = \mathbf{1}/11$, $P = I - J/11$, the remainder obeys two simultaneous lower bounds:

$$y^T R y \geq w_{\min}(1 - \rho), \quad (5)$$

$$y^T R y \geq c_0 - 2\ell \sqrt{\rho - 1/11} + e_{\min}(\rho - 1/11), \quad (6)$$

where

$$\begin{aligned} c_0 &= 0.13721547758397512, & \ell &= 0.048732847203611034, \\ e_{\min} &= -0.6703735434067706, & w_{\min} &= 0.011070817321442113. \end{aligned}$$

The center-to-remainder term uses a feasible SOCP-dual water-fill lower bound, not an unconstrained eigenvalue guess.

A 100×100 initial grid was recursively split only where its outward-paid lower bound was nonpositive. The completed tree contains 224,626 accepted leaves, 71,542 refined nodes, and maximum depth nine. The smallest lower bound after a total 10^{-10} IEEE/libm payment per evaluated cell is

$$2.437168031224723 \times 10^{-9} > 0. \quad (7)$$

It occurs in a depth-eight cell near $t \in [0.10919548, 0.10922149]$ and $\rho \in [0.11697443, 0.11700995]$.

Theorem 1 (Uniform global regime). *For the supplied strict operator (1) and every supplied $0 \leq g \leq 2.8934$, the uniform probability is the unique global minimizer of V_g on the complete simplex.*

The proof joins (4) and the complementary cover. Strictness holds in both sectors away from u , and monotonicity propagates the $g = 2.8934$ certificate to smaller nonnegative gains.

3.3 Existence, but not identity, of the first transition

For $p \neq u$, the equal-value gain is

$$R(p) = \frac{2D(p||u)}{(p-u)^T A(p-u)}. \quad (8)$$

Near u its directional lower limit is

$$\liminf_{p \rightarrow u} R(p) \geq \frac{12}{\lambda_{\max}(A)} = 5.123427551398616,$$

well above the relevant interval. Hence the infimum is attained away from u on a compact subset. Theorem 1 supplies the lower bound. The ST342 interval state has certified negative value at $g = 2.9024964917477667$, supplying the upper bound.

Theorem 2 (First global-transition bracket). *The first supplied gain at which a nonuniform state can match or beat the uniform state exists and lies in*

$$\boxed{[2.8934, 2.9024964917477667]}.$$

The bracket width is 0.0090964917477665.

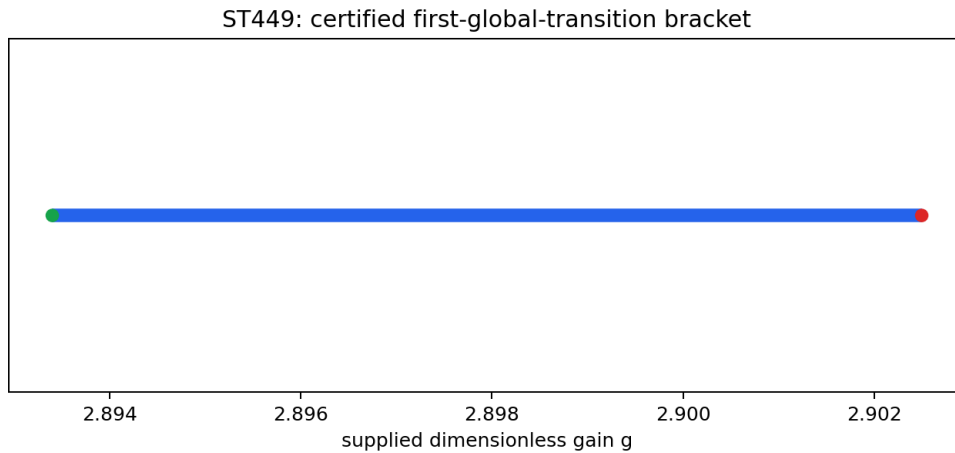


Figure 1: The certified lower regime and a certified negative competitor bound the first global transition. The figure does not identify the first orbit.

The strongest falsification safeguard is explicit: the reflection-even stationary branch crosses zero transversely near the upper endpoint, but an unknown third orbit could realize (8) earlier. Neither global branch identity nor uniqueness is claimed.

4 ST447 and ST454: exact invariant-algebra advances

4.1 Degree eight: no support-three relation

The degree-eight candidate list has

$$\binom{9}{4} + \binom{7}{2} 32 + \binom{33}{2} + 6 \cdot 117 = 2028$$

columns, while the Molien dimension is 1892, forcing nullity at least 136. Release 10.86 excluded support one and two. ST447 projects the full 64-row evaluation matrix to a collision-free projective plane over each of $\mathbb{F}_{1000003}$ and $\mathbb{F}_{1000033}$. Every projected collinear triple is then checked in the original 64-row matrix; projection may create a false candidate but cannot hide a true dependency.

The two audits enumerate all 2,055,378 column pairs. They respectively send 1,337 and 1,390 candidate triples to the full-matrix rank test. None has rank below three.

Proposition 1. *No rational relation among the declared degree-eight candidates has support at most three. Therefore every one of the at least 136 forced relations has support at least four.*

This is an exclusion theorem, not an explicit relation basis.

4.2 Degree six: exact closure in characteristic zero

The 326 decomposable candidates are

$$56 (q^3) + 192 (qP_4) + 78 (D_3D_3).$$

Column elimination at both declared primes gives rank 310 and the identical set of 16 dependent pivot columns. Chinese remaindering followed by rational reconstruction yields 16 primitive integer vectors. Their supports range from 113 to 311; the largest coefficient appearing in the packet is 512016.

Modular agreement alone would be evidence, not a characteristic-zero upper rank theorem. Therefore every lifted combination was constructed as an exact integer polynomial and divided synthetically by

$$L = x_0 + x_1 + \cdots + x_{11}.$$

All sixteen remainders are literally the empty polynomial. The relations are independent because the column-elimination form has distinct leading dependent columns. Thus rank over $\mathbb{Q}$ is at most 310. Conversely, the nonzero modular rank-310 minor is a nonzero integer minor, so the rational rank is at least 310.

Theorem 3 (Exact degree-six presentation rank). *In the mean-zero quotient, the declared decomposable degree-six family has characteristic-zero rank exactly 310 and a sixteen-dimensional relation kernel. Since the full invariant dimension is 365, the primitive quotient dimension is exactly*

$$365 - 310 = 55.$$

The complete vectors, candidate descriptors, two-prime residues, and exact division audits are stored separately in `FIN_ST454_Degree6_Integer_Relations.json`. The result closes the ST439 interval [39, 55] at its upper endpoint. It does not solve the degree-eight presentation.

5 ST450–ST453: topology and the adaptive infrared chain

5.1 What the partial Morse polynomial does and does not prove

The certified nonexhaustive stationary census from ST435 has

$$M_{\text{partial}}(t) = 13 + 42t + 30t^2.$$

ST450 records the stronger identity

$$M_{\text{partial}}(t) = 1 + (1+t)(12+30t). \quad (9)$$

The correction polynomial has nonnegative coefficients, so the census obeys the strong Morse-polynomial condition for a contractible simplex, not merely the Euler check at $t = -1$.

This is still not completeness: additional critical points may contribute another $(1+t)R(t)$ with nonnegative coefficients. Boundary strata are not covered. ST451 proves a local symmetry statement only. At each of four certified index-one representatives, the Hessian commutes with the exact stabilizer; simplicity of the negative eigenvalue makes its unstable line a one-dimensional real stabilizer representation. The two global endpoints of that line remain numerical and no Conley connection matrix is promoted.

5.2 Removing the common-box bottleneck

The prior compactification uses $b = 1/y_3$ and $a = bc$. A single common box was certified only through $b = 0.017$. ST452 replaces it by direct interval equations on cells $[b_-, b_+]$ with $b_- > 0$. The apparently ill-conditioned third stationarity equation is evaluated algebraically as

$$\frac{2c}{1 + e^{-1/b}} + \frac{2T(1 - bc)b^{-1}e^{-(T-1)/b}}{1 + e^{-T/b}} - \nu = 0.$$

For the tail integral, the rigorous inequalities

$$\frac{e^{-y}}{y+1} \leq E_1(y) \leq \frac{e^{-y}}{y}, \quad y > 0,$$

avoid catastrophic interval cancellation.

Twenty-eight width-0.003 or smaller parametric Krawczyk cells cover $[0.017, 0.1]$. Their minimum inclusion margin is 1.8024×10^{-7} and their maximum weighted contraction is 0.047294. At every one of the 28 interfaces, including $b = 0.017$, a separate tiny parametric Krawczyk box is contained in both neighboring boxes. This proves that the locally unique roots belong to one connected component rather than merely producing a list of unrelated local roots.

Theorem 4 (Connected regularized IR component). *Together with ST437, the displayed Krawczyk chain proves one locally unique root component for every $b \in [0, 0.1]$. Throughout the new part,*

$$y_1 < 0.028634 < 3.78868 < y_2 < 3.87733 < 10 \leq y_3,$$

$a \in [0.00059487, 0.00367448]$, $T > 1$, every Jacobian has negative orientation, and the local Brouwer degree is -1 .

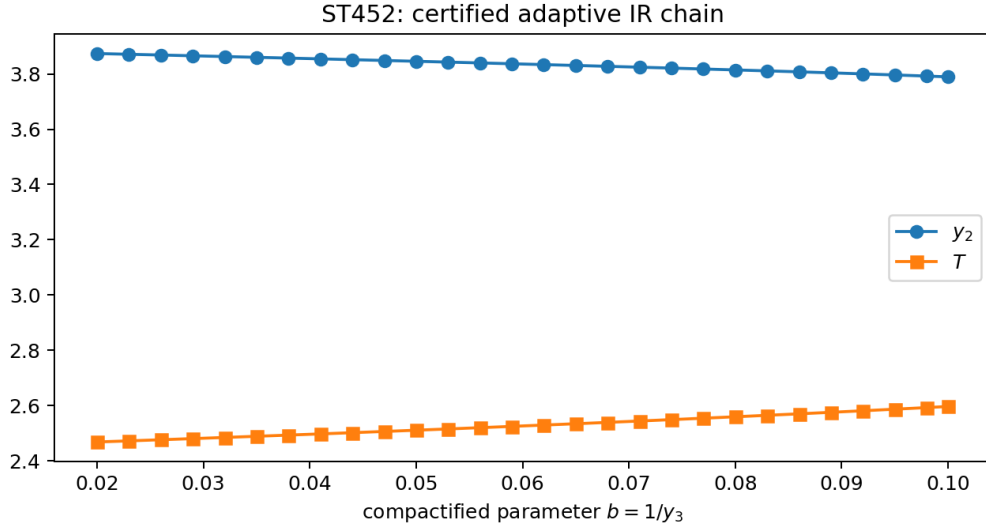


Figure 2: Centers from the certified adaptive cells. Lines are visual guides; the theorem is carried by the Krawczyk boxes and certified interface links.

No full complement cover has been performed. Disconnected positive roots outside the displayed boxes remain possible.

6 ST455–ST458: adversarial secondary results

6.1 Synthetic design: training improves, holdout worsens

Five weakest-direction row exchanges on the frozen 1,800-point synthetic pool raise the smallest singular value by a factor 1.17195 and improve the A-trace-inverse criterion by a factor 1.15870. That looks favorable if only the selected 500 rows are scored. A frozen 1,295-row holdout reverses the interpretation:

	baseline	after five exchanges
maximum leverage	3.58940	4.02581
95th percentile leverage	2.78823	2.85715

The procedure overfits its in-design weakest direction. ST455 therefore rejects the word “robust” for this exchange rule. No apparatus calibration is inferred from a synthetic pool.

6.2 Transfer sequence: the apparent plateau was premature

Extending the Ulam sequence from grids 21, 31, 41, 51, 61 to 71, 81 gives

$$\frac{|\lambda_2|}{|\lambda_1|} = 0.5957361291 \quad (n = 71), \quad 0.5957254767 \quad (n = 81).$$

The new last-two spread 1.0652×10^{-5} is larger, not smaller, than the previous 2.7505×10^{-6} spread. The sequence remains monotone over the tested grids, but visual stabilization is not an error theorem. Only the coarser analytic Birkhoff bound remains certified.

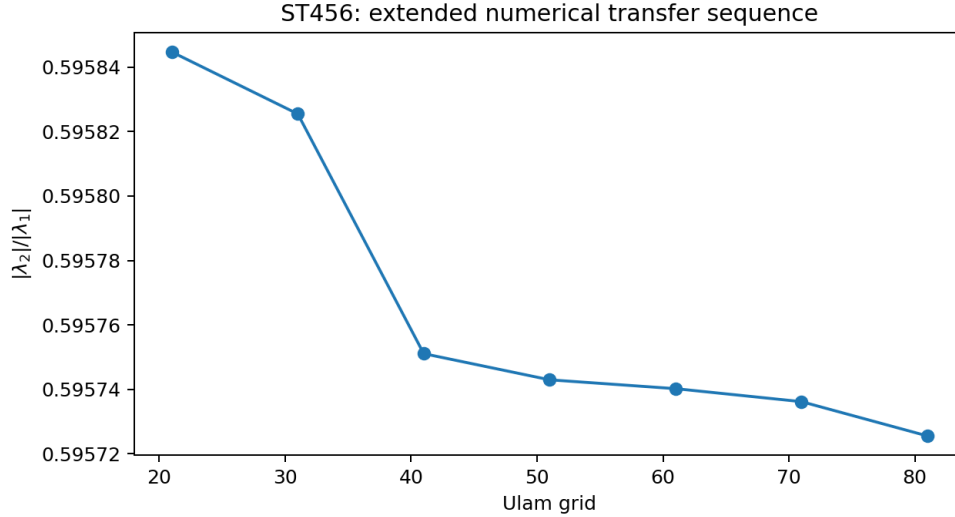


Figure 3: The numerical ratio continues to drift. No infinite-dimensional two-sided spectral-gap bracket is claimed.

6.3 A sharp result inside one conservative ISS architecture

For the previously certified local minimum, the scalar estimate is

$$\mu(R) = \mu_0 - \frac{R}{(p_{\min} - R)^2}, \quad \varepsilon(R) = \frac{R\mu(R)}{2}.$$

It has a unique interior optimum satisfying

$$\mu_0 = \frac{2Rp_{\min}}{(p_{\min} - R)^3}.$$

With $p_{\min} = 0.0002146796264$ and $\mu_0 = 86.33235115$, the result is

$$R_* = 1.9360740425 \times 10^{-6}, \quad \mu(R_*) = 43.55546688, \quad \varepsilon_* = 4.2163304418 \times 10^{-5}.$$

This proves optimality only within the stated Hessian-variation formula, not a global basin and not a physical noise law.

ST458 also retains an honest stop. The rank-seven mediator has 24 positive and 42 negative off-diagonal pairs. Positive-weight water filling does not apply, while spectral and absolute-row-sum bounds are too weak at $g = 4$. No sign-aware global SDP certificate is available, so the known local rank-seven state is not promoted to a global minimizer.

7 Falsification ledger and scientific interpretation

Claim tested	What survived	What failed or remains open
The local branch crossing is the first global transition	A first transition is now tightly bracketed.	No theorem identifies the attaining orbit; a third competitor is admissible.
Degree-six modular rank is characteristic-zero rank	Sixteen lifted relations verify exactly and close rank 310.	The same closure has not been achieved at degree eight.
Low-support degree-eight relations should be easy to find	Support one, two, and three are exactly excluded.	The at-least-136-dimensional kernel must be denser; no basis is known.

Claim tested	What survived	What failed or remains open
The IR branch stops near $b = 0.017$	The stop was a common-box artifact; adaptive boxes reach 0.1.	A global root count and further continuation are open.
More design exchanges improve robustness	In-design E and A metrics improve.	Frozen holdout leverage worsens; robustness is falsified.
The Ulam ratio had stabilized	The tested sequence remains orderly and monotone.	New grids reveal larger drift; there is no certified plateau.
Symmetry-resolved saddles determine connections	The unstable eigenline respects the stabilizer.	Global heteroclinic endpoints and a Conley matrix are not proved.
Rank-seven signs can be ignored	The local state and rank survive.	Sign-blind global envelopes remain invalid.

The global-transition theorem is a real mathematical advance for the supplied finite variational family. It is not yet a physical phase transition because g , a clock, dimensions, preparation, measurement, environment, apparatus, and record are not generated. Likewise, exact invariant relations organize the algebra of the finite state representation but do not create matter, gauge fields, gravity, or experimental observables.

8 Recommended next programs ST462–ST476

The next work should exploit the newly narrowed proof frontiers rather than repeat closed selector or generic bridge searches.

ST462: Resolve the remaining global-transition strip. Use a branch-and-bound ratio cover on $g \in [2.8934, 2.9024964917477667]$ to identify or exclude a third orbit. Highest scientific priority.

ST463: Certify the attaining orbit. Couple an interval stationary atlas directly to the ratio $R(p)$, not merely to V_g at selected gains.

ST464: Transition uniqueness and order. If one orbit survives, prove whether the first change is a unique first-order coexistence event or a multi-orbit tie.

ST465: Complete stationary-simplex exhaustion. Construct an interval cover including all boundary strata before making any complete Morse or Conley claim.

ST466: Certified connection matrix. Use isolating blocks around the four index-one unstable lines and rigorous exit/entry sets.

ST467: Degree-eight sparse kernel at support four and five. Extend projective incidence to four-column circuits using matroid/coding techniques; stop before brute-force $\binom{2028}{4}$ enumeration.

ST468: Checkpointed degree-eight nullspace. Obtain an exact modular basis at one prime, then lift only a sparse/structured subset and verify it in characteristic zero.

ST469: Minimal generator audit through degree eight. Combine the new degree-six relation module with Molien dimensions and the first lifted degree-eight relations.

ST470: Adaptive IR continuation beyond $b = 0.1$. Use variable cell width and asymptotic E_1 bounds to seek the first genuine loss of nonsingularity rather than another fixed-box stop.

ST471: Global IR complement degree. Build a bounded positive-domain exclusion cover and compare boundary degree with the local value -1 .

ST472: Holdout-aware design optimization. Optimize the maximum of training A/E loss and frozen holdout leverage; do not reuse the holdout for final reporting.

ST473: Transfer-operator discretization theorem. Derive an explicit Ulam approximation error in an adapted bounded-variation or cone norm before computing more grids.

ST474: Rank-seven signed quadratic SDP. Formulate a reproducible moment/SOS or copositive relaxation with rational dual witnesses and an explicit resource stop.

ST475: Alternative ISS Lyapunov geometries. Test entropy/Fisher and state-dependent metrics against the now-sharp scalar bound; improvement must be certified, not sampled.

ST476: Admission gates and state map. Re-run gain, selector, dimension, bridge-role-transfer, and independent-evidence gates only after a genuinely new typed source or record appears.

The recommended order is ST462, ST463, ST468, ST470, ST466, then ST472. It prioritizes the smallest finite certificates that can change theorem status.

9 Reproducibility packet

The release packet contains:

- `fin_st447_st461_research.py`: all finite computations, certificate generation, exact lifting, figures, and packet writing;
- `test_fin_st447_st461.py`: live and serialized consistency tests;
- `FIN_ST447_ST461_Results.json` and `FIN_ST447_ST461_Summary.csv`;
- fifteen per-program JSON packets;
- `FIN_ST454_Degree6_Integer_Relations.json`, containing all 326 descriptors and all 16 integer relation vectors;
- three generated figures and a SHA-256 manifest.

The executable uses deterministic seeds. The PDF records no claim of a clean working tree: the tracked baseline hash is supplemented by the explicitly listed local source packet. Reproduction should run the research script, then the test suite, regenerate the PDF twice, and finally verify the manifest.

10 Conclusion

This batch converts three earlier limitations into sharper mathematics. The global gain problem is no longer bracketed between a remote convexity bound and $g \approx 4$; its first change is confined to a width-0.0091 interval. The characteristic-zero degree-six invariant quotient is no longer an interval; its dimension is exactly 55. The infrared continuation no longer stops at the first common-box failure; one connected certified component reaches $b = 0.1$.

Equally important, falsification remains active. The first global orbit is not identified, the Morse census is not complete, the IR root count is not global, the synthetic design overfits its selected rows, and the apparent transfer-gap plateau weakens under larger grids. Gain, selector, dimensions, physical interpretation, operational preparation/measurement, and independent evidence remain separate missing objects. The mathematically strongest surviving interpretation is therefore still a finite spectral and variational information model with rigorously analyzable branches—not a completed physical theory and not a Theory of Everything.